

3.2 Linear Momentum & Conservation

Question Paper

Course	CIEA Level Physics
Section	3. Dynamics
Topic	3.2 Linear Momentum & Conservation
Difficulty	Hard

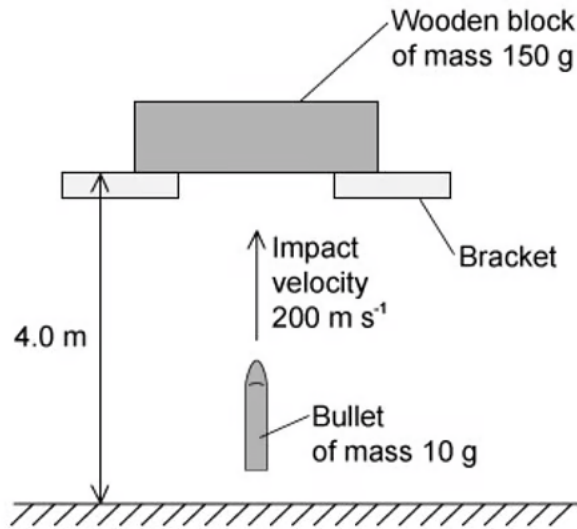
Time allowed: 20

Score: /10

Percentage: /100

Question 1

A wooden block is freely supported on brackets at a height of 4.0 m above the ground, as shown



A bullet of mass 10.0 g is shot vertically upwards into the wooden block of mass 150 g. It embeds itself in the block. The impact causes the block to rise above its supporting brackets.

The bullet hits the block with a velocity of 200 m s⁻¹. How far above the ground will the block be at the maximum height of its path?

- A. 8.0 m
- B. 10.0 m
- C. 12.0 m
- D. 14.0 m

[1 mark]

Question 2

Two railway trucks of masses m and $5m$ move towards each other in opposite directions with speeds $3u$ and u respectively. These trucks collide and stick together.

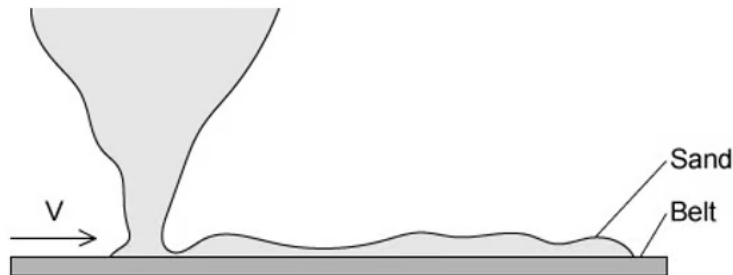
What is the speed of the trucks after the collision?

- A. $\frac{u}{6}$
- B. $\frac{3u}{6}$
- C. $\frac{u}{3}$
- D. $\frac{5u}{3}$

[1 mark]

Question 3

Sand falls vertically on a conveyor belt at a rate of $m \text{ kg s}^{-1}$



In order to keep the belt moving at constant speed v , the horizontal force that must be exerted is equal to

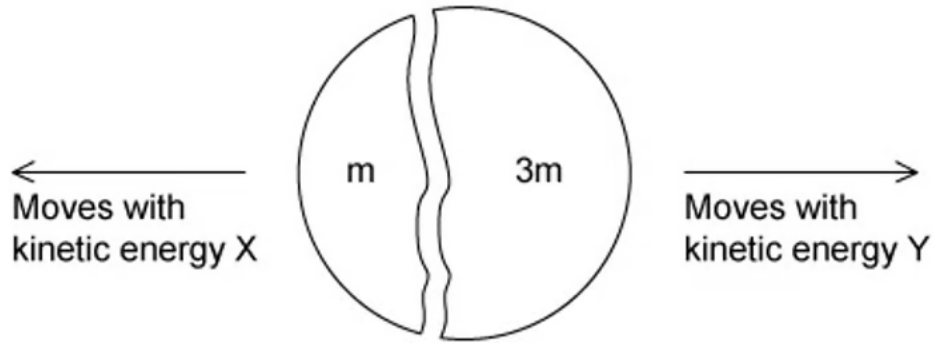
- A. mv
- B. $\frac{1}{2}mv$
- C. mv^2
- D. $\frac{1}{2}mv^2$

[1 mark]

Question 4

A stationary body explodes into two components of masses m and $3m$.

The components gain kinetic energies X and Y respectively.



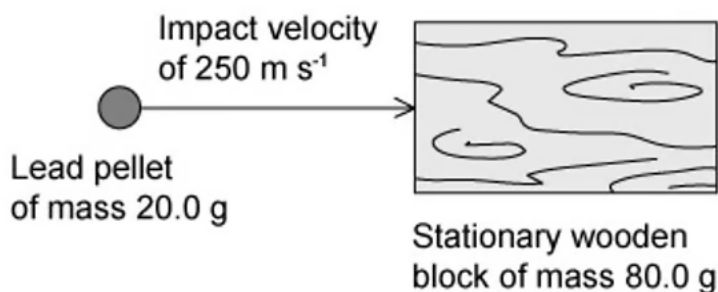
What is the value of the ratio $\frac{x}{y}$?

- A. $\frac{6}{1}$
- B. $\frac{3}{1}$
- C. $\frac{1}{3}$
- D. $\frac{1}{6}$

[1 mark]

Question 5

A lead pellet of mass 20.0 g is shot horizontally into a stationary wooden block of mass 80.0 g. The pellet hits the block with an impact velocity of 250 m s^{-1} . It embeds itself in the block and it does not emerge.



After the impact, the block and the pellet continue to move for 8.4 s, how far do they travel in this time?

- A. 175 m
- B. 210 m
- C. 260 m
- D. 525 m

[1 mark]

Question 6

A stationary nucleus has nucleon number A .

The nucleus decays by emitting a proton with speed v to form a new nucleus with speed u . The new nucleus and the proton move away from one another in opposite directions.

Which equation gives in terms of A and u ?

- A. $v = \left(\frac{A}{4} - 1 \right) u$
- B. $v = (A - 1) u$
- C. $v = Au$
- D. $v = (A + 1) u$

[1 mark]

Question 7

A particle X has speed v and collides with a stationary identical particle Y. The collision is perfectly elastic.

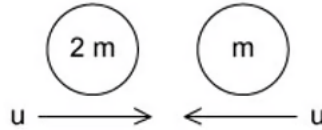
What are the speed and direction of motion of each of the two particles after the collision?

	X	Y
A	stationary	v to the right
B	$\frac{v}{2}$ to the right	$\frac{v}{2}$ to the right
C	$\frac{v}{2}$ to the left	$\frac{v}{2}$ to the right
D	v to the left	stationary

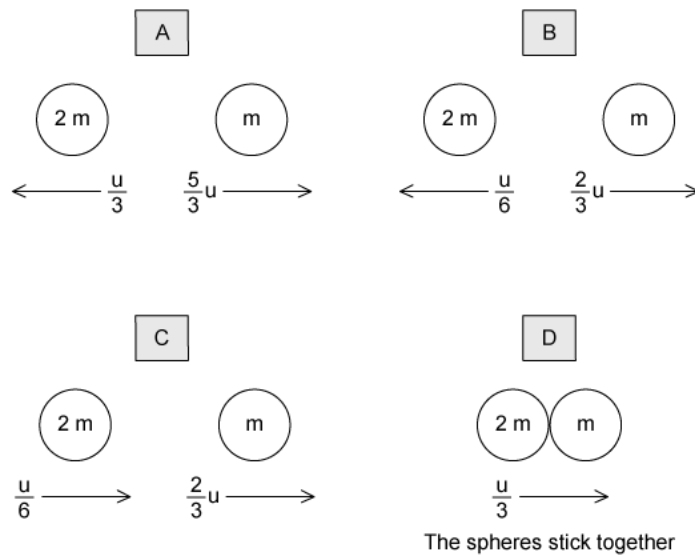
[1 mark]

Question 8

The diagram shows two spherical masses approaching each other head-on at an equal speed u . One is of mass m and the other of mass $2m$.



Which diagram, showing the situation after the collision, is not consistent with the principle of conservation of momentum?

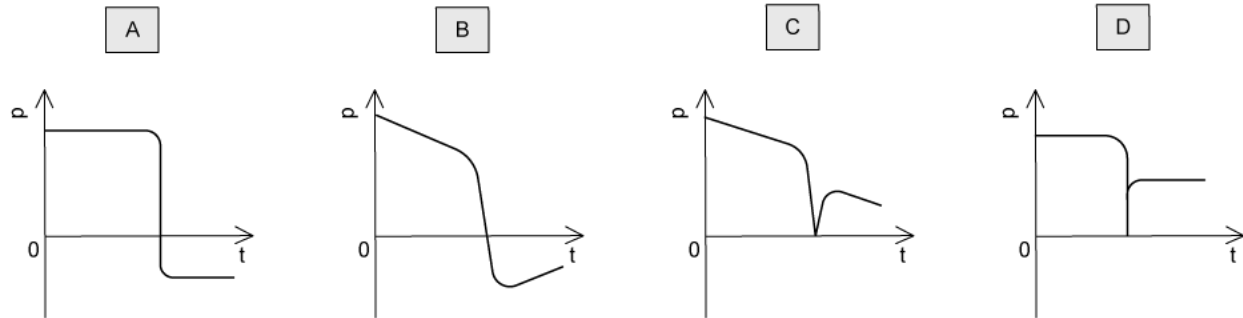


[1 mark]

Question 9

An ice-hockey puck slides along a horizontal, frictionless ice-rink surface. It collides inelastically with a wall at right angles to its path, and then rebounds along its original path.

Which graph shows the variation with time t of the momentum p of the puck?



[1 mark]

Question 10

A body, initially at rest, explodes into two masses M_1 and M_2 that move apart with speeds v_1 and v_2 respectively.

What is the ratio $\frac{v_1}{v_2}$?

- A. $\frac{M_2}{M_1}$
- B. $\frac{M_1}{M_2}$
- C. $\sqrt{\frac{M_2}{M_1}}$
- D. $\sqrt{\frac{M_1}{M_2}}$

[1 mark]

Model Answers: Hard

1

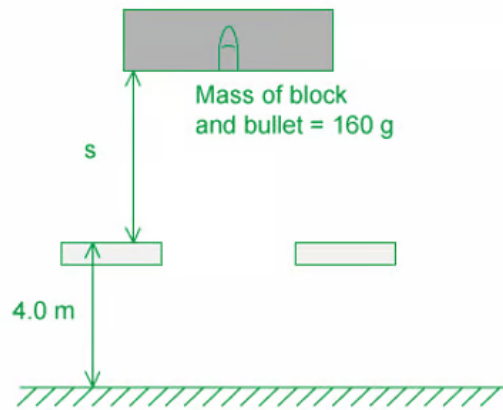
The correct answer is **C** because:

Part 1: use the conservation of momentum to find the speed after collision

- Sum of momentum before collision = Sum of momentum after collision
 - Momentum of block before collision = 0
 - Momentum of bullet before collision = $10g \times 200 \text{ m s}^{-1}$
- The bullet becomes embedded in the block, so the two objects will travel together with the same velocity, u
 - The momentum after the collision = $(10g + 150g)u = 160u$
 - $160u = 2000$
 - $u = 12.5 \text{ m s}^{-1}$

Part 2: use a kinematic equation to find the maximum height

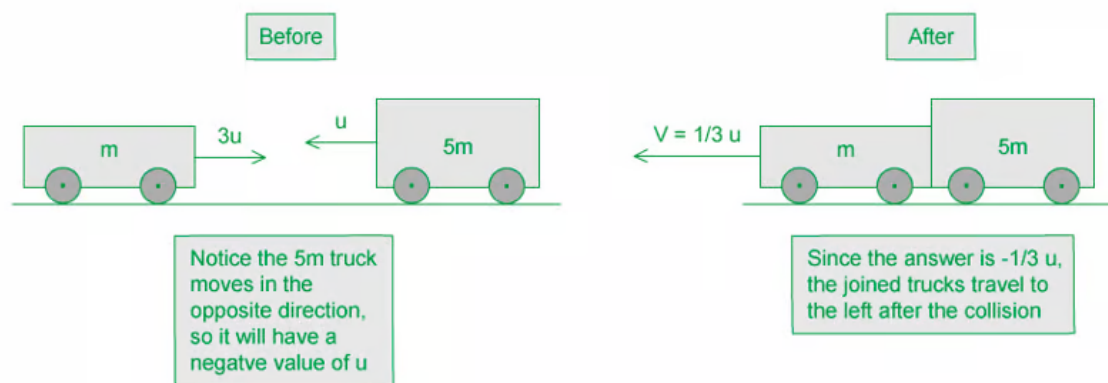
- List the quantities we know:
 - $u = 12.5 \text{ m s}^{-1}$
 - $v = 0$
 - $a = -9.81 \text{ m s}^{-2}$
 - $s = ?$
- The equation that links u , v , a and s is
 - $v^2 = u^2 + 2as$
 - $0 = 12.5^2 + 2(-9.81)s$
 - $(2 \times 9.81)s = 12.5^2$
 - $s = \frac{12.5^2}{19.6} = 8.0 \text{ m}$
- Height of the block above the ground = $4.0 + 8.0 = 12.0 \text{ m}$



2

The correct answer is **C** because:

- Using the principle of conservation of momentum:
 - Sum of momentum before collision = Sum of momentum after collision
- Let's set the positive direction to be to the right
- Momentum of trucks before collision
 - $(m \times 3u) + (5m \times -u) = 3mu - 5mu = 2mu$
- After the collision the trucks join together the total mass becomes $6m$ and the combined trucks move at a final speed v
- Momentum of trucks after collision = $(m + 5m)v = 6mv$
 - $-2mu = 6mv$
 - $u = -3v$
 - $v = -\frac{1}{3}u$
- Speed does not have a direction so the answer is $\frac{u}{3}$



3

The correct answer is **A** because:

- Usually when we deal with momentum questions it is the velocity that is changing, but in this case, the velocity is constant and it is the mass of the sand that is changing at a rate of $m \text{ kg s}^{-1}$ (m here means mass, not metres)
- We can define the rate of change of mass as
 - $m = \frac{\Delta M}{t}$
- So, the momentum is equal to
 - $\Delta p = \Delta M v$
- Force is equal to the rate of change of momentum
 - $F = \frac{\Delta p}{t} = \frac{\Delta M v}{t} = \left(\frac{\Delta M}{t} \right) v = m v$
- We can check the units to ensure it is consistent
 - The unit of force is the newton: $\text{N} = \text{kg m s}^{-2}$
 - m has units of kg s^{-1} , v has units of m s^{-1}
 - So, $m v$ also has units of $[\text{kg s}^{-1}] [\text{m s}^{-1}] = \text{kg m s}^{-2}$

B is incorrect because the units are consistent but the factor of $\frac{1}{2}$ is incorrect

C & D are incorrect because the units are inconsistent with the units of force (kg m s^{-2})
 $m v^2 = [\text{kg s}^{-1}] [\text{m s}^{-1}]^2 = \text{kg m}^2 \text{s}^{-3}$

4

The correct answer is **B** because:

- From the conservation of momentum, the sum of the momentum before the explosion should equal the sum of momentum after the explosion
 - Momentum before explosion = 0
 - Momentum after explosion = $p_{3m} - p_m$
 - $p_{3m} - p_m = 0$
 - Therefore, $p_{3m} = p_m$
- This means that the magnitudes of the momentum of both components are equal
- Let's say the velocity of mass $3m$ is v and the velocity of mass m is v'
 - Momentum of component $3m$: $p_{3m} = 3mv$
 - Momentum of component m : $p_m = mv'$
 - $3mv = mv'$
 - $v' = \frac{3mv}{m} = 3v$
- Kinetic energy = $\frac{1}{2}mv^2$
 - $X = \frac{1}{2}(m)(3v)^2 = \frac{9}{2}mv^2$
 - $Y = \frac{1}{2}(3m)(v)^2 = \frac{3}{2}mv^2$
- Ratio $\frac{X}{Y} = \frac{\frac{9}{2}mv^2}{\frac{3}{2}mv^2} = 3$

5

The correct answer is **B** because:

Part 1: use the conservation of momentum to find the speed after collision

- From the conservation of momentum, the sum of the momentum before the collision should equal the sum of momentum after the collision
 - Momentum of wooden before the collision = 0
 - Momentum of pellet before the collision = $20\text{ g} \times 250\text{ m s}^{-1}$
- After the collision the pellet becomes embedded in the wooden block, so the two

travel together with a mass of $(20\text{ g} + 80\text{ g}) = 100\text{ g}$ and a speed of u

- Momentum of block and pellet after the collision = $100\text{ g} u$
- Sum of momentum before the collision = Sum of momentum after the collision



Part 2: use a kinematic equation to find the maximum distance

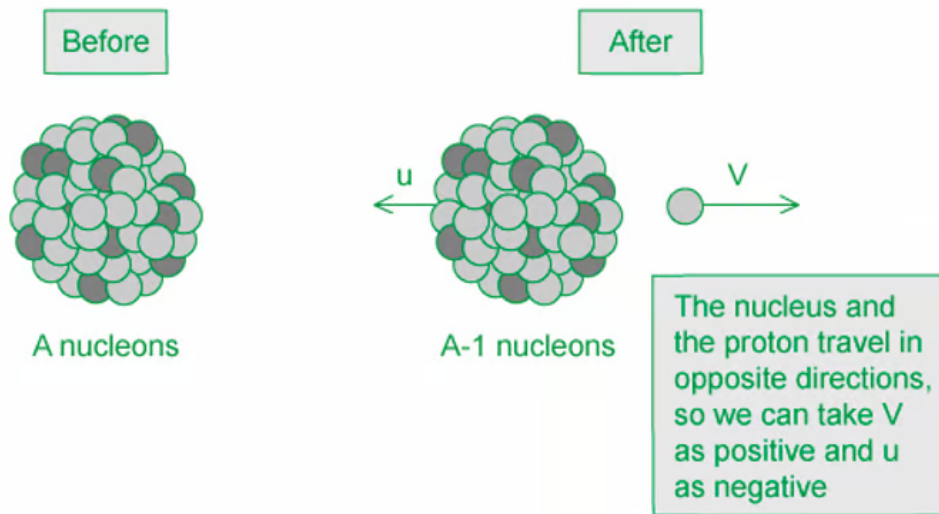
- List the quantities we know:
 - $u = 50 \text{ m s}^{-1}$
 - $v = 0$
 - $t = 8.4 \text{ s}$
 - $s = ?$
- The equation that links u , v , t and s is
 - $s = \frac{(u + v)t}{2}$
 - $s = \frac{50}{2} \times 8.4$
 - $s = 210 \text{ m}$
- The block and pellet will travel 210 m before coming to a stop

6

The correct answer is **B** because:

- According to conservation of momentum
- Sum of momentum before decay = Sum of momentum after decay
- The nucleus is initially stationary, its speed is zero, therefore, its momentum before the decay is also initially zero
- The quantities we will need to work out the momentum after the decay are:
 - The nucleus contains A nucleons
 - The proton has a speed of v
 - The new nucleus contains $A - 1$ nucleons
 - The new nucleus has a speed of u
 - Each nucleon has a mass of m
- After the nucleus decays :
- Momentum of proton = mv
- Momentum of new nucleus = $(A - 1)m(-u)$

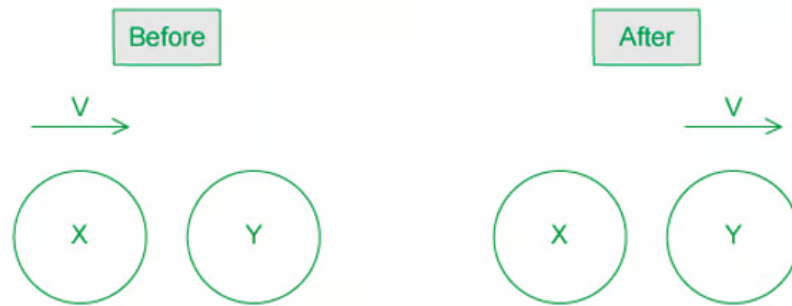
- Total momentum after decay = $mv - (A - 1)mu$
 - $0 = mv - (A - 1)mu$
 - $v = (A - 1)u$



7

The correct answer is **A** because:

- In a perfectly elastic collision, both momentum and energy are conserved
- According to conservation of momentum:
 - Sum of momentum before collision = Sum of momentum after collision
- In an elastic collision, the total kinetic energy before the collision should be the same as the total kinetic energy after
- Consider option A :
- Before collision :
 - Momentum = $mv + 0 = mv$
 - Kinetic energy = $\frac{1}{2}mv^2 + 0 = \frac{1}{2}mv^2$
- After collision:
 - Momentum = $0 + mv = mv$
 - Kinetic energy = $0 + \frac{1}{2}mv^2 = \frac{1}{2}mv^2$
- Momentum and energy are both conserved so option A is correct



B is incorrect because

$$\text{Momentum: } m\left(\frac{1}{2}v\right) + m\left(\frac{1}{2}v\right) = mv$$

$$\text{Kinetic energy: } \frac{1}{2}m\left(\frac{1}{2}v\right)^2 + \frac{1}{2}m\left(\frac{1}{2}v\right)^2 = \frac{1}{4}mv^2$$

C is incorrect because

$$\text{Momentum: } m\left(\frac{1}{2}v\right) - m\left(\frac{1}{2}v\right) = 0$$

$$\text{Kinetic energy: } \frac{1}{2}m\left(\frac{1}{2}v\right)^2 + \frac{1}{2}m\left(\frac{1}{2}v\right)^2 = \frac{1}{4}mv^2$$

D is incorrect because

$$\text{Momentum: } 0 - mv = -mv$$

$$\text{Kinetic energy: } 0 + \frac{1}{2}mv^2 = \frac{1}{2}mv^2$$

8

The correct answer is **B** because:

- According to conservation of momentum:
 - Sum of momentum before collision = Sum of momentum after collision
- Sum of momentum before collision = $2mu - mu = mu$
- Sum of momentum after collision

$$\circ \frac{2m(-u)}{6} + \frac{2mu}{3} = \frac{2mu}{3} - \frac{2mu}{6} = \frac{2mu}{3} - \frac{mu}{3} = \frac{mu}{3}$$

- The momentum before and after the collision is different so momentum has not been conserved in Option B

A is incorrect because momentum has been conserved

$$\text{sum of momentum after collision: } \frac{5mu}{3} - \frac{2mu}{3} = \frac{3mu}{3} = mu \quad \checkmark$$

C is incorrect because momentum has been conserved

$$\text{sum of momentum after collision: } \frac{2mu}{3} + \frac{2mu}{6} = \frac{2mu}{3} + \frac{mu}{3} = \frac{3mu}{3} = mu \quad \checkmark$$

D is incorrect because momentum has been conserved

$$\text{sum of momentum after collision: } \frac{(2m + m)u}{3} = \frac{3mu}{3} = mu \quad \checkmark$$

9

The correct answer is **A** because:

- The fact that the ice-rink is frictionless is significant because it means the speed of the ice-hockey puck is not affected by friction and can be assumed to travel with constant speed before the collision
- This is represented on the graph by a straight horizontal line, since momentum is equal to $p = mv$
- After the collision, the puck will rebound along its original path and its momentum will be negative as it travels in the opposite direction
- The collision is inelastic, so the puck loses some energy and will rebound with a lower speed
- On the graph, at a later time the line will be straight and horizontal but below the x-axis and will have a lower value of momentum

B & C are incorrect because the speed is constant, so the momentum is constant and this should be represented by straight, horizontal lines on the graph

D is incorrect because momentum is a vector, so if the original direction is taken as positive, the direction after the rebound is negative and should be below the x-axis

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10

The correct answer is **A** because:

- The principle of conservation of momentum can be written as
 - $M_1 u_1 + M_2 u_2 = M_1 v_1 + M_2 v_2$
- The two masses are both initially at rest so
 - $u_1 = u_2 = 0$
 - $M_1 v_1 + M_2 v_2 = 0$
- The masses move away from each other in opposite directions, so let's define v_1 as the positive direction and v_2 as the negative direction
 - $M_1 v_1 - M_2 v_2 = 0$
 - $M_1 v_1 = M_2 v_2$
 - $M_2 = \frac{v_1}{v_2} M_1$

○ $\frac{v_1}{v_2} = \frac{M_2}{M_1}$

